

DR. HUGO ROUSSEAU

Born in Lyon (France), June, 14th 1994 (31 yo)

Maison Climat Planète, IGE Grenoble, France

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EDUCATION & EXPERIENCES

- 2025-pres* **Post-doctoral researcher**
Team CryoDyn - IGE,CNRS, University Grenoble Alpes, INRAE, IRD, Grenoble
INP,Grenoble, France
Seismic investigation of subglacial hydrological processes
Supervisor: F. Gimbert
- 2022/04-
2024/06* **Post-doctoral researcher**
Department of Geography - University of Zürich, Switzerland
Alpine Mass Movements - ETH Zürich, Switzerland
*Implementation and analysis of a Material Point Method for numerical modelling of
fractures in rapid ice flow*
Supervisors: J. Gaume and M.P. Lüthi
- 2018/10-
2021/11* **PhD degree**
Grenoble-Alpes University, INRAE, UR ETNA, Grenoble, France
*From particle scale to continuum modelling of size segregation in bedload transport:
theoretical and experimental study*
Supervisors: P. Frey and J. Chauchat
- 2017–2018* **Master’s degree “Multi-phase flows, Energetics and Combustion”**
Aix-Marseille University, Marseille, France
Rank: first of the promotion
- 2015–2018* **Engineering school POLYTECH Marseille, France**
Mechanics and Energetics with major in advanced simulations
Distinction: first prize by the jury
- 2013–2015* **2-year selective preparatory classes leading to the French Grandes Écoles**
Lycée Champollion, Grenoble, France
Speciality physics and engineering science
- 2009–2013* **Student in high-school specialised in mountaineering, ski and climbing**
Moutiers, French-Alps, France
 - Baccalauréat (national high school leaving diploma) majoring in science
 - First degree of ski patrol rescuer (French national degree)

ADDITIONAL RESEARCH EXPERIENCES

- 2018/02-
2018/08 **Toward two-phase flow modelling of granular size segregation**
Internship at Laboratoire des Ecoulements Géophysique et Industriels (LEGI), Grenoble, France
Supervisor: J. Chauchat
- 2017/05-
2017/08 **Validation of a Material Point Method (MPM) model using experiments**
Internship at Ecole Polytechnique Fédérale Lausanne (EPFL), Switzerland
Supervisor: G. Rousseau and C. Ancey

PUBLICATIONS

Publications in international journals

1. Wehrlé, A., Lüthi, M.P., Kneib-Walter, A., Nap, A., **Rousseau, H.**, Juvet, G., & Walter, F.. Velocity and calving response of a major Greenland ice stream to a lake drainage event. *Nat. Geosci.* (2025). <https://doi.org/10.1038/s41561-025-01858-2>
2. Wehrlé, A., **Rousseau, H.**, Lüthi, M. P., Nap, A., Kneib-Walter, A., van Ginkel, J., Juvet, G. & Walter, F. (2025). Immediate ice stream and shear margin response to calving at Sermeq Kujalleq in Kangia, Greenland. *Geophysical Research Letters*, 52(18), <https://doi.org/10.1029/2025GL115829>.
3. Nap, A., Hudson, T., Walter, F., Wehrlé, A., Kneib-Walter, A., **Rousseau, H.**, & Lüthi, M. P. (2025). Interpreting ice anisotropy of Greenland's fastest outlet glacier using basal icequakes. *Geophysical Research Letters*, 52(12), <https://doi.org/10.1029/2025GL116225>.
4. Dedieu, B., **Rousseau, H.**, Chauchat, J., & Frey, P. (2024). Exploring the size ratio impact on an intruder segregating in bedload transport. *Physical Review Fluids*, 9(10), 104302. <https://doi.org/10.1103/PhysRevFluids.9.104302>
5. **Rousseau, H.**, Gaume, J., Blatny, L., & Lüthi, M. P. (2024). Transition between mechanical and geometric controls in glacier crevassing processes. *Geophysical Research Letters*, 51(9), e2024GL108206. <https://doi.org/10.1029/2024GL108206>.
6. Rousseau, G., Métivet, T., **Rousseau, H.**, Daviet, G., & Bertails-Descoubes, F. (2023). Revisiting the role of friction coefficients in granular collapses: confrontation of 3-D non-smooth simulations with experiments. *Journal of Fluid Mechanics*, 975, A14, <https://doi.org/10.1017/jfm.2023.835>.
7. **Rousseau, H.**, Chauchat, J. & Frey, P. Experiments on a single large particle segregating in bedload transport. *Phys. Rev. Fluids* **7**(6), 064305. <https://doi.org/10.1103/PhysRevFluids.7.064305>.
8. **Rousseau, H.**, Chassagne, R., Chauchat, J., Maurin, R., & Frey, P. (2021). Bridging the gap between particle-scale forces and continuum modelling of size segregation: Application to bedload transport. *Journal of Fluid Mechanics*, 916, A26. <https://doi.org/10.1017/jfm.2021.218>.

Communications in international and national conferences

1. Rousseau, H., Gaume, J., Blatny, L., and Lüthi, M. P.: Modelling discontinuities in ice flow using the Material Point Method and elastoplasticity, **EGU General Assembly 2023**, Vienna, Austria, 24–28 Apr 2023, EGU23-11460, <https://doi.org/10.5194/egusphere-egu23-11460>, 2023.

2. Wehrlé, A., Lüthi, M. P., Nap, A., Kneib-Walter, A., Jouvét, G., Rousseau, H., and Walter, F.: Calving response to the propagation of a speedup pulse through the ice stream of Sermeq Kujalleq in Kangia (Jakoshavn Isbræ), Greenland, **EGU General Assembly 2023**, Vienna, Austria, 24–28 Apr 2023, EGU23-11695, <https://doi.org/10.5194/egusphere-egu23-11695>, 2023.
3. Rousseau, H., Chauchat, J. & Frey, P. Combining theory, numeric and experiments to model size segregation in bedload transport. **Journée Mech’Alps Jeunes Chercheurs**, col de Porte, Grenoble, France, May 4th & 5th, 2022.
4. Rousseau, H., Chauchat, J. & Frey, P. Experimental investigation of a large particle segregating in bedload transport, 25th **International Congress of Theoretical and Applied Mechanics (ICTAM)**, Milano, Italy, August 22th-27th, 2021.
5. Rousseau, H., Chassagne, R., Chauchat, J., Maurin, R., & Frey, P. Continuous modelling of grain-size segregation in bedload transport, **EGU General Assembly Conference**, Vienna, Austria, May 4th-8th, 2020.
6. Rousseau, H., Chassagne, R., Chauchat, J., & Frey, P. Continuous modelling of grain-size segregation in bedload transport. **Winter school-Multi scale approaches and multiphysics couplings in fluid and solid mechanics**, Grenoble, France, January 20th-20th, 2020. (Poster session)
7. Frey, P., Rousseau, H., Chassagne, R., Maurin, R., & Chauchat, J. Sediment transport : from discrete element methods to continuum models, **AGU Fall Meeting**, San Francisco, CA, December 9th-13th, 2019. (Poster session)
8. Rousseau, H., Chauchat, J., & Frey, P. Toward Eulerian-Eulerian two-phase flow modelling of grain-size segregation in bedload transport, **Two-pHase modeling for Sediment dynamIcS (THESIS)**, University of Delaware, USA, September 18th 2019.
9. Rousseau, H., Chauchat, J., & Frey, P. Modelling grain-size segregation in sediment transport using a Eulerien-Eulerian two-phase flow model, 3rd **IMA conference on Dense Granular Flows**, Cambridge, England, July 2nd 2019.

REVIEWING ACTIVITIES

- Reviewer for the international journal *Journal of Geophysical Research - Earth Surface*
- Reviewer for the international journal *The Cryosphere*
- Reviewer for the international journal *Journal of Fluid Mechanics*

TEACHING ACTIVITIES

- 2024–2025* Teaching Granular Mechanics at ETH Zürich:
- *Lectures on grain size-segregation (3 hours)*
- Teaching at Grenoble-Alpes University (UGA):
- *Lectures (in english) on free surface flows, Master Environmental Fluid Mechanics (9 hours)*
- 2023–2024* Teaching Granular Mechanics at ETH Zürich:
- *Lectures on grain size-segregation (4.5 hours)*
 - *Tutorials on RockyDEM a software for discrete element simulations (3 hours)*
- 2018–2021* Teaching at INP Grenoble engineering school for water, energy and environment (ENSE³):
- *Tutorials in numerical simulations with OpenFOAM (12 hours)*
 - *Laboratories on sediment transport (64 hours)*
- Teaching at Grenoble-Alpes University (UGA):
- *Lectures (in english) on free surface flows, Master Environmental Fluid Mechanics (12 hours)*

STUDENTS SUPERVISION

- 2024* **Undergraduate students** (6 weeks) - Baptiste GUILLEMINOT - 3rd year at Magistère de Physique Paris-Saclay
- Toward an emulator for Alpine mass movements based on physics informed deep learning*

COMPUTING SKILLS

- Programming in various languages: Python, Fortran, C++, Matlab, L^AT_EX
- Knowledge of RockyDEM, a software dedicated to Discrete Element Modelling of Granular Materials
- Knowledge of Houdini FX, a software for 3D visualization, rendering and animation
- Knowledge of basic Python libraries for deep learning: GNNs and PINNs using Pytorch and CNN using tensorflow

POPULARIZATION ACTIONS

- 2019*
- Presentation of experiments on snow and granular avalanches at “Les Tribulations savantes” to students from 6 to 14 years old.
 - Presentation of experiments on snow and granular avalanches at “la Fête de la science” to students and interested peoples.
- 2017-2018* Creation of an organization named “Pôle Avenir” to organize conferences at the engineering school Polytech Marseille.

LANGUAGES

French: Mother tongue

English: C1 level

Spanish: School level

OTHER INTERESTS

- Alpinism, rock-climbing, ski, paragliding
- Guitar and Drama